

BUBAI MAJI

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PROFILE

Final-year Ph.D. student at IIT Kharagpur specializing in **speech and audio signal processing**, with a focus on **speech enhancement, robust speech representation learning, and multimodal audio modeling**.

My research emphasizes **perceptual audio quality, noise-robust modeling, and cross-lingual generalization**, with applications in real-time speech systems. I am particularly interested in developing **scalable, low-latency, and immersive audio technologies**.

AUDIO & SIGNAL PROCESSING EXPERTISE

- Speech enhancement (single-channel, noise-robust modeling)
- Time-frequency analysis (STFT, Mel-spectrograms)
- Self-supervised speech models (wav2vec2, HuBERT)
- Audio pipelines (Librosa, TorchAudio, FFmpeg)
- Multimodal audio-text modeling (cross-modal transformers)
- Cross-lingual and robust speech modeling
- Speech dataset design and annotation

EDUCATION

Ph.D. in Speech & Language Processing, IIT Kharagpur

Research Topic: Speech enhancement, emotion-aware modeling, depression detection

Advisors: Prof. [Rajlakshmi Guha](#) and Prof. [Aurobinda Routray](#)

Tentative Thesis Submission: November 2026

August 2022 – present

West Bengal, India

MTech, Indian Institute of Engineering Science and Technology Shibpur (GPA: 8.33/10.0)

Department of Electronics and Telecommunication Engineering

July 2018 – August 2020

West Bengal, India

BTech, MAKAUT University (GPA: 8.21/10.0)

Department of Electronics and Communication Engineering

July 2013 – June 2017

West Bengal, India

RESEARCH & PROJECT EXPERIENCE

Noise-Robust Speech Enhancement using Dual-Memory xLSTM

- Designed a single-channel speech enhancement model for noisy environments
- Focused on noise-aware representation learning
- Targeted improved perceptual speech quality

Multimodal Emotion-Aware Speech Modeling

- Built cross-modal transformer models integrating audio and text
- Developed temporal feature learning for emotion recognition

Self-Supervised Speech Representation Learning

- Investigated layer-wise representations in wav2vec2 and HuBERT
- Studied robustness across languages and acoustic conditions

Mobile & Web-based Speech Systems

- Built end-to-end pipeline: data collection, annotation, modeling

- Developed Android and web-based applications

PUBLICATIONS (SELECTED)

Speech & Audio Processing

- NGDM-SE: Noise-Guided Dual-Memory xLSTM for Single-Channel Speech Enhancement (Submitted in **INTERSPEECH 2026**)
- Investigation of Layer-Wise Speech Representations in Self-Supervised Learning Models: A Cross-Lingual Study in Detecting Depression **INTERSPEECH 2024** [\[PDF\]](#)
- Exploring Self-Supervised Models for Depressive Disorder Detection: A Study on Speech Corpora **IEEE EMBC 2024** [\[PDF\]](#)

Multimodal & Speech Modeling

- Multimodal Emotion Recognition Based on Deep Temporal Features Using Cross-Modal Transformer and Self-Attention **ICASSP 2023** [\[PDF\]](#)
- Cross-Lingual Speech Emotion Recognition: A Comparative Study Between Indo-Aryan and Western Language Families (Submitted in **INTERSPEECH 2026**)

Advanced Speech Applications

- Towards Fair ASR For Second Language Speakers Using Fairness Prompted Finetuning **ICASSP 2026** [\[PDF\]](#)
- A Study on The Impact of Foundation Models on Automatic Depression Detection from Speech Signals **INTERSPEECH 2025** [\[PDF\]](#)
- Prosody Disentanglement with Self-Supervised Speech Representation for Detecting Depression **ICASSP 2025 Workshop** [\[PDF\]](#)

TECHNICAL SKILLS

Programming: Python, C++, Java

Frameworks: PyTorch, Hugging Face Transformers

Audio Tools: Librosa, Praat, TorchAudio, FFmpeg

Systems: Docker, Git, Linux, Slurm

Other: OpenCV, Streamlit, LaTeX

PROFESSIONAL ACTIVITIES

Reviewer:

- IEEE Transactions on Computational Social Systems
- IEEE Transactions on Affective Computing
- Biomedical Signal Processing and Control, Elsevier
- Computer Methods in Biomechanics and Biomedical Engineering
- Scientific Reports
- ICASSP